# LINES THROUGH AT LEAST TWO POINTS OF AN $8 \times n$ GRID: A PROOF OF BARKER'S ORDER-20 RECURRENCE (OEIS A160848)

MATH0

ABSTRACT. Let $a(n)$ be the number of lines in the plane that pass through at least two points of the $8 \times n$ grid of lattice points (OEIS A160848). We prove the conjecture of Colin Barker (2017) that $a$ satisfies, for all $n > 21$, the linear recurrence of order 20 with characteristic polynomial $(1-x)^3\Phi_2\Phi_3\Phi_4\Phi_5\Phi_6\Phi_7$, namely $a(n) = -a(n-2)+a(n-5)+\cdots+a(n-18)+a(n-20)$. The proof shows that the second difference $a(n+1)-2a(n)+a(n-1)$ is, for $n \geq 3$, an integer combination of gcd indicators with periods at most 7; this is a rigorous form, for eight rows, of a recursion that S. Mustonen found experimentally in 2009 and supported by a heuristic argument. We found no earlier proof of Barker's recurrence and classify the result as new, though elementary. The proof is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definition of $a$ is checked by the kernel against all 41 listed terms.

## 1. INTRODUCTION

For $n \geq 0$ let $\Gamma_n = \{0, \ldots, n-1\} \times \{0, \ldots, 7\} \subset \mathbb{Z}^2$, and let $a(n)$ be the number of lines in $\mathbb{R}^2$ containing at least two points of $\Gamma_n$. This is A160848 of the OEIS [7], "Number of lines through at least 2 points of an 8 X n grid of points", contributed by Seppo Mustonen on May 28, 2009, with offset 0 and first terms $0, 1, 66, 131, 238, 361, 534, \ldots$; it is column 8 of the array A295707 [8]. On December 25, 2017, Colin Barker added to the entry a conjectured rational generating function and the corresponding linear recurrence of order 20 (quoted in Section 2). In revision #13 (December 25, 2017), which is still the current revision of the entry in October 2026, both are labelled "Conjectures". We prove the recurrence (Theorem 1), and with it the generating function (Corollary 2).

**Prior work.**

- Mustonen [6] proves a closed formula $L(m,n) = \frac{1}{2}\big(f(m,n,1)-f(m,n,2)\big)$ for the number $L(m,n)$ of lines through at least two points of an $m \times n$ grid, by summing over slopes (his Section 3); it is the formula quoted in A160848. In his Appendix 2 he observes experimentally that for fixed $m$ the residual $L(m,n) - 2L(m,n-1) + L(m,n-2)$ is periodic in $n$, and gives it explicitly in his formula (38), supported by an explanation in terms of the first two grid points of a line (his "bearing points") and by numerical checks. The Mathematica program in A160848 implements (38).
- Haukkanen and Merikoski [3], Remark 2, write that for rectangular grids with unequal sides "the question about recursive formulas is more difficult. Mustonen [5] conjectured a recursive formula for $l_{\geq 2}(n_1, n_2)$", where $l_{\geq 2}(n_1, n_2)$ is the number of lines through at least two points of an $n_1 \times n_2$ grid.
- Ernvall-Hytönen, Matomäki, Haukkanen and Merikoski [1] prove Mustonen's conjectured recursive formulas for the $n \times n$ and $(n-1) \times n$ grids; they do not treat grids with a fixed number of rows.

The math0 referees found no proof or refutation of Barker's recurrence for A160848; the analogous formulas in the sibling entries A160845–A160847, A160849 and A160850 are likewise still

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labelled conjectural or empirical. Our Proposition 5 is Mustonen's formula (38) for $m = 8$ (Remark 6); once it is proved, the recurrence follows from the periodicity of the residual by a routine cyclotomic argument. We therefore regard the result as new but elementary: a first rigorous proof of a consequence of a formula that was previously supported only heuristically.

## 2. Statement

**Theorem 1.** *For every integer $n > 21$,*

$$\begin{aligned} a(n) = {} & -a(n-2) + a(n-5) + a(n-6) + 2a(n-7) + a(n-8) + a(n-9) \\ & - a(n-11) - a(n-12) - 2a(n-13) - a(n-14) - a(n-15) + a(n-18) + a(n-20). \end{aligned}$$

*Equivalently, $\sum_{k=0}^{20} p_k\, a(n-k) = 0$ for $n \geq 22$, where*

$$P(x) = \sum_{k=0}^{20} p_k x^k = 1 + x^2 - x^5 - x^6 - 2x^7 - x^8 - x^9 + x^{11} + x^{12} + 2x^{13} + x^{14} + x^{15} - x^{18} - x^{20}.$$

Expanding shows $P(x) = (1-x)^3(1+x)(1-x+x^2)(1+x^2)(1+x+x^2)(1+\cdots+x^4)(1+\cdots+x^6) = (1-x)^3 \prod_{d=2}^{7} \Phi_d(x)$, which is Barker's denominator. The recurrence fails for $n = 20$ and $n = 21$ (the left side minus the right side is 8 in both cases), so the range $n > 21$ is sharp.

**Corollary 2** (not formalized)**.** *$\sum_{n\geq 0} a(n)x^n = xN(x)/P(x)$, where $xN(x)$ is the numerator in Barker's generating function below.*

*Proof.* The coefficient of $x^n$ in $P(x)\sum_n a(n)x^n$ is $\sum_{k\leq \min(n,20)} p_k a(n-k)$. For $n \geq 22$ it vanishes by Theorem 1, as does the coefficient of $x^n$ in $xN(x)$, which has degree 21. For $n \leq 21$ the two coefficients agree by a direct computation from $a(0), \ldots, a(21)$, which are among the kernel-checked terms (`a_data` below). □

**The OEIS conjecture.** The FORMULA section of A160848 [7] contains, verbatim:

```
Conjectures from _Colin Barker_, Dec 25 2017: (Start)
G.f.: x*(1 + 66*x + 132*x^2 + 304*x^3 + 492*x^4 + 771*x^5 + 1003*x^6 + 1273*x^7 + 1390*x^8 + 1480*x^9 +
    1374*x^10 + 1241*x^11 + 971*x^12 + 739*x^13 + 476*x^14 + 296*x^15 + 140*x^16 + 74*x^17 + 17*x^18 + 8*x
    ^19 + 8*x^20) / ((1 - x)^3*(1 + x)*(1 - x + x^2)*(1 + x^2)*(1 + x + x^2)*(1 + x + x^2 + x^3 + x^4)*(1
    + x + x^2 + x^3 + x^4 + x^5 + x^6)).
a(n) = -a(n-2) + a(n-5) + a(n-6) + 2*a(n-7) + a(n-8) + a(n-9) - a(n-11) - a(n-12) - 2*a(n-13) - a(n-14) - a
    (n-15) + a(n-18) + a(n-20) for n>21.
(End)
```

**Lean formalization.** The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A160848.lean`, reviewed for fidelity and frozen (SHA-256 `8883f318...633f3`) before any proof attempt. The definitions on which the statement depends, and the statement itself, read as follows (verbatim and in order; the header, import, docstrings, the definitions used only for fast evaluation and the data check, the theorems, the `Decidable` instance for `Recurrence` and the unproved variant `statement_gf` are omitted):

```
namespace Math0.Statements.A160848
def pt (i : ℕ) : ℤ × ℤ := (((i / 8 : ℕ) : ℤ), ((i % 8 : ℕ) : ℤ))
def Collinear3 (p q r : ℤ × ℤ) : Prop :=
  (q.1 - p.1) * (r.2 - p.2) = (q.2 - p.2) * (r.1 - p.1)
instance (p q r : ℤ × ℤ) : Decidable (Collinear3 p q r) :=
  inferInstanceAs (Decidable (_ = _))
def lineThrough (n i j : ℕ) : Finset ℕ :=
  (Finset.range (8 * n)).filter fun k => Collinear3 (pt i) (pt j) (pt k)
def a (n : ℕ) : ℕ :=
```

```
  (((Finset.range (8 * n) ×ˢ Finset.range (8 * n)).filter fun ij => ij.1 ≠ ij.2).image
    fun ij => lineThrough n ij.1 ij.2).card
def Recurrence (f : ℕ → ℤ) (n : ℕ) : Prop :=
  f n = -f (n - 2) + f (n - 5) + f (n - 6) + 2 * f (n - 7) + f (n - 8) + f (n - 9)
    - f (n - 11) - f (n - 12) - 2 * f (n - 13) - f (n - 14) - f (n - 15) + f (n - 18) + f (n -
    20)
def statement : Prop :=
  ∀ n : ℕ, 21 < n → Recurrence (fun m => (a m : ℤ)) n
end Math0.Statements.A160848
```

The point with number $i < 8n$ is $\mathrm{pt}(i) = (\lfloor i/8 \rfloor, i \bmod 8)$, so $i \mapsto \mathrm{pt}(i)$ is a bijection from $\{0, \dots, 8n-1\}$ onto $\Gamma_n$ and the number of $(x, y)$ is $\nu(x, y) = 8x + y$. (OEIS writes "$8 \times n$"; $\Gamma_n$ and its transpose are congruent, so they have the same number of lines.) For $p \neq q$, `Collinear3`$(p, q, r)$ says that $r$ lies on the line $pq$. Thus `lineThrough`$(n, i, j)$ is the set of grid points on the line through $\mathrm{pt}(i) \neq \mathrm{pt}(j)$, and `a`$(n)$ is the number of distinct such sets. A line through two grid points is determined by its set of grid points, and two distinct lines share at most one point, so `a`$(n)$ is exactly the number of lines through at least two points of $\Gamma_n$. The values are cast to $\mathbb{Z}$ because of the negative coefficients; for $n > 21$ every index $n-k$ ($k \leq 20$) is at least 2, so the truncated subtractions in $\mathbb{N}$ are exact. The statement file also proves the following (statements shown, proofs omitted); `IsFirst`, `goodF` and `aFast` are explained in Lemma 3:

```
theorem isFirst_iff {i j : ℕ} (hij : i < j) : IsFirst i j ↔ goodF i j = true
theorem a_eq_aFast (n : ℕ) : a n = aFast n
theorem a_data : (List.range 41).map a =
    [0, 1, 66, 131, 238, 361, 534, 709, 938, 1183, 1470, 1759, 2104, 2459, 2870, 3287, 3740,
    4209,
      4734, 5261, 5844, 6437, 7070, 7711, 8408, 9115, 9872, 10637, 11444, 12265, 13142, 14015,
      14944, 15889, 16876, 17871, 18914, 19967, 21076, 22193, 23352]
```

The list in `a_data` is the full DATA section of the entry, $a(0), \dots, a(40)$; it is proved by `decide +kernel` after rewriting with `a_eq_aFast`. This anchors the definition to the entry independently of the conjecture. The file also checks that these terms satisfy the recurrence for $22 \leq n \leq 40$ (`recurrence_data`), a check of the transcription only.

## 3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A160848.lean`; bracketed names are the corresponding Lean declarations. Write $[\,A\,]$ for the indicator of a statement $A$ and $u(a, b) = [\,\gcd(a, b) = 1\,]$ for $a, b \in \mathbb{N}$; note $u(a, 0) = [\,a = 1\,]$.

### 3.1. Counting each line once (statement file).

Call a pair $i < j < 8n$ *first* (`IsFirst`) if no $k < j$ with $k \neq i$ has $\mathrm{pt}(k)$ on the line through $\mathrm{pt}(i)$ and $\mathrm{pt}(j)$, that is, if $i, j$ are the two smallest elements of `lineThrough`$(n, i, j)$.

**Lemma 3** (`a_eq_card_firstPairs`, `isFirst_iff`, `a_eq_aFast`)**.** *$a(n)$ is the number of first pairs. For $i < j$ write $\mathrm{pt}(i) = (x, y)$, $\mathrm{pt}(j) = (x', y')$ and $d = (x' - x, y' - y)$. Then $(i, j)$ is first if and only if $\gcd(x' - x, |y' - y|) = 1$ and the point $\mathrm{pt}(i) - d = (2x - x', 2y - y')$ does not lie in $\mathbb{N} \times \{0, \dots, 7\}$, i.e. not ($x' \leq 2x$ and $y' \leq 2y < 8 + y'$). This is the test `goodF`$(i, j)$, and `aFast`$(n)$, the number of pairs $i < j < 8n$ passing it, equals $a(n)$.*

*Proof.* If $\mathrm{pt}(i) \neq \mathrm{pt}(j)$ both lie on the line through $p \neq q$, then $r$ lies on the line $\mathrm{pt}(i)\mathrm{pt}(j)$ iff it lies on $pq$ (`collinear_iff`, a polynomial identity). Hence sending a set $S =$ `lineThrough`$(n, p, q)$ to its two smallest elements is inverse to $(i, j) \mapsto$ `lineThrough`$(n, i, j)$ on first pairs, which proves the first claim.

The numbering is additive: $\nu(P+w)=\nu(P)+8w_1+w_2$. Put $P=\mathrm{pt}(i)$, so $j-i=8d_1+d_2>0$ and $d_1\ge 0$. If $g=\gcd(d_1,|d_2|)\ge 2$, the point $P+d/g$ lies in $\Gamma_n$ (its coordinates lie between those of $\mathrm{pt}(i)$ and $\mathrm{pt}(j)$), on the line, with number $i+(j-i)/g\in(i,j)$. If $P-d\in\mathbb{N}\times\{0,\dots,7\}$, then also $x-d_1\le x<n$, so $P-d\in\Gamma_n$, on the line, with number $2i-j<i$. Either way $(i,j)$ is not first. Conversely, let $\gcd(d_1,|d_2|)=1$, $P-d\notin\mathbb{N}\times\{0,\dots,7\}$, and suppose $\mathrm{pt}(k)=P+e$ is on the line with $k<j$, $k\ne i$, i.e. $d_1e_2=d_2e_1$. With $\alpha d_1+\beta d_2=1$ (Bézout) and $t=\alpha e_1+\beta e_2\in\mathbb{Z}$ we get $e=td$, hence $k-i=t(j-i)$, and $k<j$, $k\ne i$ force $t\le -1$. Since $-1$ lies between $t$ and 0, the point $P-d$ lies on the segment from $P+td$ to $P$, both of which are in the convex set $[0,\infty)\times[0,7]$; as $P-d\in\mathbb{Z}^2$, it follows that $P-d\in\mathbb{N}\times\{0,\dots,7\}$ (in Lean: coordinatewise inequalities), a contradiction. □

3.2. **Adding a column.** Put $D(n)=a(n+1)-a(n)$, the number of good pairs $(i,j)$ (good meaning `goodF`$(i,j)$) with $\mathrm{pt}(j)$ in column $n$ [`aFast_succ`]. For $x<n$ let $C(n,x)$ be the number of good pairs with $\mathrm{pt}(i)$ in column $x$ and $\mathrm{pt}(j)$ in column $n$. With $\mathrm{pt}(i)=(x,y)$, $\mathrm{pt}(j)=(n,t)$ and $x\le n$, Lemma 3 says that the pair is good iff

$$\gcd(n-x,|t-y|)=1 \quad\text{and not}\quad \big(n\le 2x \text{ and } t\le 2y<8+t\big) \tag{1}$$

[`goodF_eq`]. Define

$$G(a)=8u(a,0)+\sum_{b=1}^{7}2(8-b)\,u(a,b),\qquad G^{\flat}(a)=8u(a,0)+12u(a,1)+8u(a,2)+4u(a,3).$$

**Lemma 4** (`diag_part`, `D_eq`, `C_eq`)**.** *$D(n)=1+\sum_{x<n}C(n,x)$ and $C(n,x)=G(n-x)-[\,n\le 2x\,]\,G^{\flat}(n-x)$ for $x<n$.*

*Proof.* Inside column $n$ ($x=n$), (1) requires $|t-y|=1$, so $t=y+1$, and then excludes every $y\ge 1$; the only good pair is $(8n,8n+1)$, which gives the vertical line $x=n$. For $x<n$, group the 64 row pairs $(y,t)$ by $b=|t-y|$. There are 8 pairs with $b=0$ and $2(8-b)$ with $1\le b\le 7$; this gives $G$. The rows $t$, $y$, $2y-t$ form an arithmetic progression with difference $\pm b$ and middle term $y$, so the number of pairs with $|t-y|=b$ and $0\le 2y-t<8$ is $8,12,8,4$ for $b=0,1,2,3$ and 0 for $b\ge 4$; this gives $G^{\flat}$. (In Lean the 64 terms are expanded and collected by `simp` and `ring`.) □

3.3. **The second difference.** Let $h(m)=G(m)-[\,m \text{ even},\ m\ge 2\,]\,G^{\flat}(m/2)$ [`hcl`].

**Proposition 5** (`C_succ_sub`, `sum_odd`, `D_succ`, `second_diff`)**.** *For every $m\ge 0$, $a(m+2)-2a(m+1)+a(m)=h(m+1)$.*

*Proof.* By Lemma 4, $D(n+1)-D(n)=C(n+1,0)+\sum_{x<n}\big(C(n+1,x+1)-C(n,x)\big)$, and $C(n+1,0)=G(n+1)$. Shifting both columns by one keeps $n-x$ fixed, and $[\,n+1\le 2x+2\,]-[\,n\le 2x\,]=[\,n=2x+1\,]$, so $C(n+1,x+1)-C(n,x)=-[\,n=2x+1\,]G^{\flat}(n-x)$. The only candidate is $x=(n-1)/2$ for odd $n$, where $n-x=(n+1)/2$. Hence $D(n+1)-D(n)=h(n+1)$, and $a(m+2)-2a(m+1)+a(m)=D(m+1)-D(m)$. □

**Remark 6** (Mustonen's formula (38))**.** In [6], $\psi_1(m,n)=\sum_{1\le y<m,\,(y,n-1)=1}(m-y)$ and, for odd $n$, $\psi_4(m,n)=\sum_{1\le y<m,\,((n-1)/2,y)=1}\psi_3(m,n,\frac{n-1}{2},y)$, where $\psi_3$ is the second difference in $n$ of $(n-x)(m-y)-(n-2x)^+(m-2y)^+$; $\psi_4=0$ for even $n$. For $m=8$, $2\psi_1(8,n)=G(n-1)-8u(n-1,0)$, and for odd $n=2k+1$ one finds $\psi_3(8,n,k,y)=-(8-2y)^+$, so $2\psi_4(8,n)=-(G^{\flat}(k)-8u(k,0))$. For $n\ge 4$ the terms with $u(\cdot,0)$ vanish, so Proposition 5, with $n-2$ in place of $m$, is (38) for eight rows, where $L(8,n)=a(n)$. Lemma 3 makes rigorous the role of Mustonen's "bearing points".

3.4. **Periodicity.**

**Lemma 7** (`u_zero`, `G_period`, `Gb_period`, `hcl_period`)**.** *For $m \geq 3$,*

$$h(m) = \sum_{b=1}^{7} 2(8-b)\,u(m,b) - [\,2 \mid m\,]\big(12 + 8\,u(m/2,2) + 4\,u(m/2,3)\big), \tag{2}$$

*and $h(m) = h(m \bmod 420 + 420)$.*

*Proof.* For $m \geq 2$, $u(m,0) = [\,m = 1\,] = 0$, and for even $m \geq 3$, $m/2 \geq 2$ and $u(m/2,0) = 0$; this gives (2). Every $b \in \{1,\dots,7\}$ divides 420, so $u(m \bmod 420 + 420, b) = u(m,b)$, because $\gcd(a,b)$ depends only on $a \bmod b$. For even $m$, $(m \bmod 420 + 420)/2 = (m/2) \bmod 210 + 210$, and $2, 3$ divide 210. □

3.5. **The kernel certificate.** Let $\hat{h}(m) = h(m \bmod 420 + 420)$ for $m \in \mathbb{N}$ [`hp`], and let

$$R(x) = \sum_{j=0}^{18} r_j x^j = 1 + 2x + 4x^2 + 6x^3 + 8x^4 + 9x^5 + 9x^6 + 7x^7 + 4x^8 \\ - 4x^{10} - 7x^{11} - 9x^{12} - 9x^{13} - 8x^{14} - 6x^{15} - 4x^{16} - 2x^{17} - x^{18}.$$

Expanding, $P(x) = (1-x)^2 R(x)$ and $R(x) = (1-x)\prod_{d=2}^{7} \Phi_d(x)$.

**Lemma 8** (`key`)**.** *For every $s \in \{0,\dots,419\}$, $\sum_{j=0}^{18} r_j\,\hat{h}(s+21-j) = 0$.*

*Kernel proof.* The Lean statement, $\forall s < 420,\ \hat{h}(s+21) + 2\hat{h}(s+20) + \cdots - \hat{h}(s+3) = 0$ in $\mathbb{Z}$, is decidable, and `decide +kernel` has the kernel itself reduce the decision procedure to `true`: it evaluates $420 \cdot 18$ values of $\hat{h}$, each a few gcds of numbers below 840 with numbers $b \leq 7$, using the kernel's built-in arithmetic on natural-number literals. No compiled code is trusted and no axiom is added; the certificate is the evaluation itself. □

*Conceptual proof.* By Lemma 7, (2) holds for $\hat{h}$ at every $m \in \mathbb{N}$, since $m \bmod 420 + 420 \geq 3$ is congruent to $m$ modulo 420. Hence $\hat{h}$ is a $\mathbb{Z}$-linear combination of the sequences $u(m,b)$ $(1 \leq b \leq 7$; period $b)$, $[\,2 \mid m\,]$ (period 2), $[\,m \equiv 2 \bmod 4\,] = [\,2 \mid m\,]u(m/2,2)$ (period 4) and $[\,m \equiv 2,4 \bmod 6\,] = [\,2 \mid m\,]u(m/2,3)$ (period 6). For $q \leq 7$, $1-x^q = (1-x)\prod_{1<d\mid q} \Phi_d(x)$ divides $R$, say $R = (1-x^q)S_q$. For a $q$-periodic $f$, $\sum_j r_j f(M-j) = \sum_i s_{q,i}\big(f(M-i) - f(M-i-q)\big) = 0$ whenever $M \geq 18$. Take $M = s+21$. □

3.6. **Proof of Theorem 1.** Let $n = N+22$, $s = N \bmod 420$, and $\Delta^2 a(m) = a(m+2) - 2a(m+1) + a(m)$. Since $P(x) = (1-x)^2 R(x) = \sum_j r_j(x^j - 2x^{j+1} + x^{j+2})$,

$$\sum_{k=0}^{20} p_k\,a(n-k) = \sum_{j=0}^{18} r_j\,\Delta^2 a(N+20-j).$$

By Proposition 5, $\Delta^2 a(N+20-j) = h(N+21-j)$, and since $N+21-j \geq 3$, Lemma 7 gives $h(N+21-j) = \hat{h}(s+21-j)$ [`hcl_eq`]. The right-hand side is therefore $\sum_j r_j \hat{h}(s+21-j) = 0$ by Lemma 8. In Lean, `result` rewrites the indices $N+22-k$, instantiates Proposition 5 at the 18 indices with $r_j \neq 0$, and closes the goal by `linear_combination` with coefficients $r_j$ and Lemma 8 at $s$. □

**Remark 9** (other numbers of rows; not formalized)**.** For $m$ rows the same steps give, for $k \geq 3$, $h_m(k) = \sum_{1\leq b<m} 2(m-b)\,u(k,b) - [\,2 \mid k\,]\sum_{1\leq b<m/2} 2(m-2b)\,u(k/2,b)$, a combination of sequences of periods at most $m-1$, and the number of lines satisfies the recurrence with characteristic polynomial $(1-x)^3\prod_{d=2}^{m-1}\Phi_d(x)$, of degree $2+\sum_{d<m}\varphi(d)$, for $n \geq 4+\sum_{d<m}\varphi(d)$. We checked this numerically for $2 \leq m \leq 9$; we have not compared it with the sibling entries.

## 4. Machine verification

**Files and declarations.** The proof is `lean/Math0/Frontier/OEIS/A160848.lean` in the math0 repository, `https://github.com/Foundation-0/math0` (299 lines). It imports the frozen statement file `lean/Math0/Statements/OEIS/A160848.lean` and the audit commands `lean/Math0/Audit.lean`. The only public declaration is `Math0.Frontier.A160848.result : Math0.Statements.A160848.statement`; the auxiliary lemmas of this file named in Section 3 are private. The kernel certificate is the private lemma `key` (Lemma 8; its statement lists the 18 nonzero coefficients $r_j$ explicitly and its proof is `decide +kernel`), and the file ends with the audit command `#math0_settles Math0.Frontier.A160848.result Math0.Statements.A160848.statement`. Neither file contains `sorry`, `native_decide`, custom axioms or `maxHeartbeats`; the only option is `maxRecDepth` on `key` (and on `a_data` in the statement file).

**Audit.** The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata, with no unfolding. Here it reports that `statement` and `result` each depend on the axioms `[Classical.choice, Quot.sound, propext]`, and ends with

```
math0 statement check: 'Math0.Frontier.A160848.result' proves 'Math0.Statements.A160848.statement'
```

**Toolchain and reproduction.** We use Lean 4 [5] `v4.33.1` and Mathlib [4] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use, since the statement is a math0 formalization of an OEIS conjecture. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, runs `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file is checked by `lake env lean Math0/Frontier/OEIS/A160848.lean` in `lean`, in under two minutes on our machine (about one minute of CPU time), including the kernel evaluation of Lemma 8; it prints the two audit lines and the statement check quoted above.

## References


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